

5G: The Next-Generation NETWORK



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With time, generations change, whether it is mankind or technological. But the biggest and fastest changes have only been noticed in the communication network generations in the last few years. We have noticed a great electronic change in wireless communication network generations ranging from 1G to 2G, 3G and 4G in a very short span.

Traditionally, there are three ways the mobile industry can add more capacity to its network: by adding more spectrum, by improving spectrum efficiency or by rolling out more infrastructure. Each generation is characterised by new frequency bands, higher data rates and non-backward-compatible transmission technology.

Presently, wireless networking devices have occupied a significant place in society due to their easy access and expanding utility. Such devices are almost surpassing the human number on the Earth, and it

is unimaginable to think about a place on Earth where these technologies have not reached. People now expect a lot more applications from mobile networks, and scientists and engineers along with industries are always working to make it happen.

1G refers to the first generation of wireless telephone technology (mobile telecommunications). These analogue telecommunications standards were introduced in the 1980s and continued until being replaced by 2G (second-generation wireless telephone technology) digital telecommunications.

The main difference between 1G and 2G is that the radio signals used by 1G networks are analogue, while 2G networks are digital. Although both use digital signalling to connect radio towers (which listen to handsets) to the rest of the telephone system, voice itself during a call is encoded to digital signals in 2G, whereas in 1G it is only modulated to higher frequency, typically 150MHz and up. The inherent advantages of digital technology over that of analogue meant that 2G networks eventually replaced these almost everywhere.

Second-generation, or 2G, mobile telecom networks were commercially launched in 1991. Three primary benefits of 2G networks over their predecessors were that, phone conversations were digitally encrypted, 2G systems were significantly more efficient on the spectrum, allowing far greater mobile phone penetration levels, and 2G introduced data services for mobile, starting with SMS text messages.

2G technologies enabled various mobile phone networks to provide services such as text, picture and multimedia messages (MMSes). All text messages sent over 2G are digitally encrypted, allowing for transfer of data in such a way that only the intended receiver can receive and read it.

2G has been superseded by newer

